NATURAL LANGUAGE PROCESSING

LABORATORY ASSIGNMENT – 03

Survey and Implementation of POS Tagging

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Academic Year	2026-2027
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Assignment 3: Survey and Implementation of POS Tagging

Aim: To study various techniques for Part-of-Speech (POS) tagging and implement any one POS-tagging technique using Python and suitable NLP libraries.

1. Objectives

- Understand the purpose and importance of POS tagging in Natural Language Processing.
- Study rule-based, statistical and machine-learning approaches to POS tagging.
- Understand the role of training data and tagsets in POS tagging.
- Implement one selected POS-tagging technique.
- Evaluate the implementation using suitable test data and metrics.

2. Problem Statement

Conduct a technical survey of different POS-tagging techniques and implement any one of them using Python. Use a suitable corpus or dataset, tag unseen sentences, evaluate the results where a gold-standard corpus is available, and discuss the strengths and limitations of the selected approach.

3. Theory / Background

Part-of-Speech tagging assigns a grammatical category such as noun, verb, adjective, adverb, pronoun or preposition to each word in a sentence. POS tagging is useful for parsing, information extraction, question answering, sentiment analysis and other NLP applications.

Common POS-tagging approaches to be surveyed:

- Rule-based POS tagging
- Statistical POS tagging using Hidden Markov Models (HMM)
- Transformation-based tagging
- Maximum Entropy / discriminative models
- Conditional Random Fields (CRF)
- Neural POS tagging and contextual language models

4. Survey of POS Tagging Techniques

Technique	Basic Principle	Advantages	Limitations	Reference
Rule-based	Dictionary gives candidate tags; hand-written contextual rules eliminate wrong ones.	Transparent; no training data needed; easy to debug.	Costly to write and maintain; poor coverage; language-specific.	Greene & Rubin, 1971 (TAGGIT)
HMM / Statistical	Tags are hidden states; transition P(t|t-1) and emission P(w|t) are learned from a tagged corpus; Viterbi decodes.	Simple, fast, needs only counts; good accuracy (about 95%) with small data.	Weak on unknown words; only local (n-gram) context; independence assumptions.	Rabiner, 1989 (Proc. IEEE)

Transformation-based	Start with most-frequent tags, then learn an ordered list of error-correcting rewrite rules.	Human-readable rules; compact model; accuracy comparable to HMM.	Slow training; greedy rule ordering; rule templates are hand-designed.	Brill, 1995 (Comput. Linguistics)
Maximum Entropy	Discriminative log-linear model P(t | context) using rich features (neighbouring words, prefixes, suffixes, capitalisation).	Uses arbitrary overlapping features; better OOV handling; about 96.6% on PTB.	Tags predicted left-to-right, so label-bias and error propagation; needs feature engineering.	Ratnaparkhi, 1996 (EMNLP)
CRF	Discriminative model of the whole tag sequence P(y | x) with a global normalisation over the sentence.	Avoids label bias; globally optimal sequence; flexible features.	Heavier training; manual features; limited to linear-chain context.	Lafferty et al., 2001 (ICML)
Neural / Contextual	BiLSTM or Transformer models (e.g. BERT) learn embeddings and contextual representations end-to-end.	State-of-the-art accuracy (97% or higher); handles OOV via character/sub-word units; uses full context.	Needs more data and compute; less interpretable; pre-trained models are large.	Plank et al., 2016 (ACL); Devlin et al., 2019 (NAACL)

5. Selected Technique

Technique selected for implementation: Statistical POS tagging - Hidden Markov Model (HMM) with Viterbi decoding

Reason for selection:

An HMM tagger is a classic, mathematically clean supervised approach: tags are hidden states, words are observations, and the best tag sequence is found exactly using dynamic programming (Viterbi).

It needs only simple counts from a tagged corpus, trains in seconds, is fully interpretable (transition and emission tables), and models tag-to-tag context, so it already beats a most-frequent-tag baseline.

It directly covers the viva topics for this assignment (HMM, Viterbi, smoothing, unknown words), and it is a good stepping stone to CRF and neural taggers.

6. Tools, Libraries and Environment

Sr. No.	Tool / Library	Version	Purpose
1	Python	3.12.3	Programming language for the implementation
2	NLTK	3.10.3	Corpus loading (Penn Treebank), tag mapping, tokenisation
3	scikit-learn	1.8.0	Accuracy, precision, recall, F1, confusion matrix

4	NumPy	2.4.4	Log-probability arrays and Viterbi dynamic-programming tables

5	Matplotlib	3.10.8	Confusion-matrix and per-tag score plots
6	Jupyter / VS Code	Any recent	Editor / environment for running pos_tagging_hmm.py

7. Dataset / Corpus

Corpus / Dataset Name: Penn Treebank sample (NLTK treebank corpus), plus 24 custom demonstration sentences

Source: NLTK Data repository (nltk.corpus.treebank), a ~5% sample of the Wall Street Journal portion of the Penn Treebank. The 24 demonstration sentences were written manually for this assignment.

Language: English

Number of sentences: 24 custom sentences for demonstration tagging. The model is trained and evaluated on the Treebank sample of 3,914 sentences (3,131 train / 783 test).

Tagset used: Universal POS tagset (12 tags: ADJ, ADP, ADV, CONJ, DET, NOUN, NUM, PRON, PRT, VERB, X, and. for punctuation)

Description:
The model is trained on manually POS-tagged English sentences from Wall Street Journal news articles. Original Penn Treebank tags were mapped to the Universal tagset using NLTK's map_tag. Trace elements (-NONE-) were removed and the sentences were shuffled and split 80/20 (seed 42). The 24 custom sentences are unseen, general-domain sentences covering nouns, verbs, adjectives, adverbs, numbers, pronouns and punctuation. They were tagged by the trained HMM to show its behaviour outside the news domain. They have no gold tags, so accuracy figures come only from the 783-sentence Treebank test set.

8. Methodology / Algorithm

Explain the selected POS-tagging algorithm step by step.

1. Load the tagged corpus, remove trace tokens, map PTB tags to Universal tags, shuffle (seed 42) and split $80/20$.
2. Count transitions $C(t(i-1), t(i))$ including a $< s >$ start and $< /s >$ end marker, and emissions C (tag, word) on lower-cased words.
3. Estimate transition probabilities $P(t_i \mid t_(i-1))$ with add-k smoothing ($k = 0.001$) so no tag bigram has zero probability.
4. Estimate emission probabilities $P(w \mid t) = C(t, w) \;/\; C(t)$ by maximum likelihood; all scores are kept in log space to avoid underflow.
5. Handle unknown (OOV) words: numbers are tagged NUM; otherwise use a suffix model (last 1-3 letters learned from rare training words, count <= 2), plus a boost to NOUN for capitalised words.
6. Viterbi decoding: $V[0][j] = log\ P(t_j \mid < s >) + log\ P(w_0 \mid t_j)$; $V[i][j] = max_m\ (\ V[i-1][m] + log\ P(t_j \mid t_m)\ ) + log\ P(w_i \mid t_j)$; keep back-pointers.
7. Terminate with the best final state (including the $< /s >$ transition) and follow back-pointers to recover the best tag sequence.

8. Evaluate on the test set: accuracy, precision, recall, $F1$ per tag, OOV accuracy, confusion matrix, and compare with a most-frequent-tag baseline.

9. Training / Configuration Details

Parameter	Value	Remarks
Training corpus / split	3131 sentences (80%)	Random shuffle of NLTK treebank sample
Testing corpus / split	783 sentences, 18396 tokens (20%)	Unseen during training; trace tokens removed
Tagset	Universal, 12 tags	Mapped from PTB tags with NLTK
Model parameters	Bigram HMM, $add-k\ k = 0.001$, suffix length 3	Rare-word threshold: count <= 2 (suffix model)
Random state / seed	42	random.seed(42), np.random.seed(42)
Other	Log-space Viterbi; lower-cased vocabulary	OOV: digits to NUM, suffix model, capitalised word boosts NOUN

10. Implementation / Source Code

Attached the complete source code separately.

11. Sample POS-Tagged Output

Sr. No.	Input Sentence	POS-Tagged Output	Remarks
1	The quick brown fox jumps over the lazy dog .	The/DET quick/ADJ brown/NOUN fox/NOUN jumps/NOUN over/ADP the/DET lazy/ADJ dog/NOUN ./.	'brown' (should be ADJ) and 'jumps' (should be VERB) wrongly tagged NOUN; rest correct.
2	She sells sea shells by the sea shore .	She/PRON sells/VERB sea/NOUN shells/NOUN by/ADP the/DET sea/NOUN shore/VERB ./.	'shore' wrongly tagged VERB (should be NOUN); 'sells' correct.
3	Pune is a beautiful city in Maharashtra .	Pune/NOUN is/VERB a/DET beautiful/ADJ city/NOUN in/ADP Maharashtra/NOUN ./.	All plausible; proper nouns Pune / Maharashtra appear as NOUN (Universal tagset has no PROPN).
4	I will book a flight to Delhi tomorrow .	I/PRON will/VERB book/NOUN a/DET flight/NOUN to/PRT Delhi/NOUN tomorrow/NOUN ./.	'book' wrongly NOUN after 'will' (should be VERB); 'to' tagged PRT as per Universal tagset.
5	Please read the book that she wrote .	Please/NOUN read/VERB the/DET book/NOUN that/ADP she/PRON wrote/VERB ./.	'Please' wrongly NOUN (should be VERB-like) and 'that' tagged ADP instead of DET; rest correct.
6	The company reported strong profits in 2023 .	The/DET company/NOUN reported/VERB strong/ADJ profits/NOUN in/ADP 2023/NUM ./.	Fully correct, including 2023 -> NUM (handled by digit rule).
7	Students are learning natural language processing quickly .	Students/NOUN are/VERB learning/NOUN natural/ADJ language/NOUN processing/NOUN quickly/ADV ./.	'learning' tagged NOUN (should be VERB); other tags correct.
8	The government announced new economic policies yesterday .	The/DET government/NOUN announced/VERB new/ADJ economic/ADJ policies/NOUN yesterday/NOUN ./.	All tags correct.

9	He runs five miles every morning before breakfast .	He/PRON runs/VERB five/NUM miles/NOUN every/DET morning/NOUN before/ADP breakfast/NOUN ./.	Fully correct; five -> NUM.
10	The stock market fell sharply after the report .	The/DET stock/NOUN market/NOUN fell/VERB sharply/ADV after/ADP the/DET report/NOUN ./.	Fully correct.
11	Mary gave her brother a gift for his birthday .	Mary/NOUN gave/VERB her/PRON brother/VERB a/DET gift/NOUN for/ADP his/PRON birthday/NOUN ./.	'brother' wrongly tagged VERB (should be NOUN); 'her' and 'his' are PRON in the Universal tagset.
12	Scientists have discovered a new species of frog .	Scientists/NOUN have/VERB discovered/VERB a/DET new/ADJ species/NOUN of/ADP frog/NOUN ./.	Fully correct.
13	The committee will meet on Friday to discuss the budget .	The/DET committee/NOUN will/VERB meet/VERB on/ADP Friday/NOUN to/PRT discuss/VERB the/DET budget/NOUN ./.	Fully correct; 'to' is PRT before a verb.
14	They are building a modern hospital near the river .	They/PRON are/VERB building/VERB a/DET modern/ADJ hospital/NOUN near/ADP the/DET river/NOUN ./.	Fully correct.
15	My sister works for a large software company in Bangalore .	My/PRON sister/NOUN works/NOUN for/ADP a/DET large/ADJ software/NOUN company/NOUN in/ADP Bangalore/NOUN ./.	'works' wrongly NOUN (should be VERB); the noun/verb ambiguity of 'works' is not resolved.
16	The weather was cold and the roads were slippery .	The/DET weather/NOUN was/VERB cold/NOUN and/CONJ the/DET roads/NOUN were/VERB slippery/NOUN ./.	'cold' and 'slippery' wrongly NOUN (should be ADJ): predicate adjectives, a NOUN/ADJ confusion.
17	Investors expect interest rates to rise next quarter .	Investors/NOUN expect/VERB interest/NOUN rates/NOUN to/PRT rise/VERB next/ADJ quarter/NOUN ./.	Fully correct.
18	Can you help me find the nearest railway station ?	Can/VERB you/PRON help/VERB me/PRON find/VERB the/DET nearest/ADJ railway/NOUN station/NOUN ?/.	Fully correct; modal 'Can' maps to VERB and '?' to punctuation.
19	The teacher asked the students to submit their assignments .	The/DET teacher/NOUN asked/VERB the/DET students/NOUN to/PRT submit/VERB their/PRON assignments/NOUN ./.	Fully correct.
20	Several banks have reduced their lending rates .	Several/ADJ banks/NOUN have/VERB reduced/VERB their/PRON lending/VERB rates/NOUN ./.	'lending' tagged VERB; as a noun modifier it is arguably NOUN/ADJ (debatable); rest correct.
21	The old man walked slowly across the crowded street .	The/DET old/ADJ man/NOUN walked/VERB slowly/ADV across/ADP the/DET crowded/VERB street/NOUN ./.	'crowded' wrongly VERB (should be ADJ); a participle used as an adjective.
22	Microsoft and Google are investing heavily in artificial intelligence .	Microsoft/NOUN and/CONJ Google/NOUN are/VERB investing/VERB heavily/ADV in/ADP artificial/ADJ intelligence/NOUN ./.	Fully correct; company names are NOUN (no PROPN in the Universal tagset).

23	We can not ignore the effects of climate change .	We/PRON can/VERB not/ADV ignore/VERB the/DET effects/NOUN of/ADP climate/NOUN change/NOUN ./.	Fully correct; 'not' is ADV.
24	The new law will take effect in January .	The/DET new/ADJ law/NOUN will/VERB take/VERB effect/NOUN in/ADP January/NOUN ./.	Fully correct.

12. Evaluation

Evaluation metric(s) used: Accuracy, Precision, Recall, F1-score (weighted and macro), OOV accuracy, confusion matrix

Metric	Value	Expected / Baseline	Remarks
Accuracy	0.9517	0.9259 (most-frequent-tag)	+2.58 points over baseline
Precision	0.9515	n/a	Weighted average (macro: 0.8726)
Recall	0.9517	n/a	Weighted average (macro: 0.8734)
F1-score	0.9515	n/a	Weighted average (macro: 0.8729, lowered by rare tag X)
Other	OOV accuracy 0.7946	n/a	1305 OOV tokens (7.09% of test tokens)

13. Error Analysis

Identify incorrectly tagged words/sentences and explain possible causes.

Input / Word	Predicted Tag	Correct Tag	Reason / Analysis
confident (…Drexel remains confident of its future…)	NOUN	ADJ	Unknown word ending in -ent; suffix model prefers NOUN.
supermarket (…rampage through supermarket aisles…)	ADJ	NOUN	Noun used as a modifier before another noun looks like an adjective in context.
CLEARS (INQUIRY CLEARS TEXAS JUDGE)	NOUN	VERB	Upper-case headline; capitalised OOV word is boosted toward NOUN.
that (…conversations that went nowhere…)	ADP	DET	'that' is ambiguous (DET, ADP/complementiser, pronoun); a bigram context cannot separate them.
please (…sellers please go away…)	NOUN	VERB	Rare imperative use; word seen mostly in other roles in training.
following (The following individuals were fined)	ADJ	VERB	Participle used as a determiner-like modifier; tag annotation is itself debatable.

14. Results / Output Screenshots

```
(base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 2$ cd "/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3"
(base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ pwd
/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3
(base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ find . -maxdepth 3 -type f -printf '%p (%k KB)\n' | sort
./build_notebook.py (16 KB)
./custom_predictions.csv (20 KB)
./custom_test_sentences.txt (4 KB)
./error_analysis.csv (84 KB)
./evaluation_results.csv (4 KB)
./generate_report.py (72 KB)
./generate_workbook.py (20 KB)
./.gitignore (4 KB)
./outputs/POS_Tagging_Assignment_03_executed.ipynb (612 KB)
./outputs/terminal_output.txt (4 KB)
./POS_Tagging_Assignment_03.ipynb (612 KB)
./POS_Tagging_Assignment_03_Report.docx (420 KB)
./pos_tagging_hmm.py (64 KB)
./POS_Tagging_Results.xlsx (32 KB)
./README.md (8 KB)
./requirements.txt (4 KB)
./results.json (120 KB)
./RUN_COMMANDS.txt (4 KB)
./sample_tagged_output.csv (12 KB)
./SCREENSHOT_GUIDE.md (12 KB)
./test_sentences.txt (104 KB)
./tests/test_pos_tagging.py (12 KB)
./train_sentences.txt (408 KB)
./.venv/bin/activate (4 KB)
./.venv/bin/activate.csh (4 KB)
./.venv/bin/activate.fish (4 KB)
./.venv/bin/Activate.ps1 (12 KB)
./.venv/bin/pip3.13 (4 KB)
./.venv/bin/pip3 (4 KB)
./.venv/bin/pip (4 KB)
./.venv/.gitignore (4 KB)
./.venv/pyvenv.cfg (4 KB)
./visualizations/confusion_matrix.png (132 KB)
./visualizations/model_comparison.png (60 KB)
./visualizations/oov_analysis.png (56 KB)
./visualizations/per_tag_scores.png (80 KB)
./visualizations/tag_distribution.png (80 KB)
(base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python3 --version
Python 3.13.9
(base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python3 -m pip --version
pip 25.3 from /home/shravanikurkute/anaconda3/lib/python3.13/site-packages/pip (python 3.13)
(base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ source .venv/bin/activate
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python -c "import nltk, numpy, pandas, sklearn, matplotlib; print('All core NLP libraries imported successfully')"
```

```
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python -c "from nltk.corpus import treebank; print('Treebank sentences:', len(treebank.tagged_sents()))"
Treebank sentences: 3914
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ ls -lh pos_tagging_hmm.py
-rw-r--r--. 1 shravanikurkute shravanikurkute 64K Oct 19 21:40 pos_tagging_hmm.py
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ mkdir -p outputs visualizations screenshots tests
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ set -o pipefail
python -u pos_tagging_hmm.py 2>&1 | tee outputs/terminal_output.txt
================================================================================
NLP LABORATORY ASSIGNMENT 3 - HMM PART-OF-SPEECH TAGGING
================================================================================
[resources] All required NLTK resources are available.
[corpus] Raw PTB sentences: 3,914; raw tokens: 100,676
[corpus] Trace (-NONE-) tokens removed: 6,592
[corpus] Exact duplicate sentences removed: 11
[corpus] Sentences after cleaning: 3,903
[split] Seed=42, test_size=0.20 -> 3,122 train / 781 test sentences
[split] 75,349 train tokens / 18,678 test tokens; train vocabulary=10,625; test OOV=1,467 (7.85%)
[split] Sentence overlap between train and test: 0
[custom] Loaded 25 manually written test sentences.
[train] Training the HMM ...
[train] HMM tags=12, vocabulary=10,625, emission pairs=11,271
[train] OOV suffix model built from 8,929 rare-word occurrences; numeric rare tokens=747
[predict] Decoding the held-out test set with Viterbi ...
--------------------------------------------------------------------------------
Baseline accuracy          : 0.932594
Baseline macro / weighted F1: 0.856378 / 0.931416
HMM accuracy               : 0.953475
HMM macro / weighted F1     : 0.916128 / 0.953004
HMM OOV accuracy           : 0.755965 on 1,467 OOV tokens
HMM known-token accuracy   : 0.970310 on 17,211 known tokens
Total HMM token errors     : 869
Pipeline time              : 2.33 seconds
--------------------------------------------------------------------------------
[workbook] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Results.xlsx
[report] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Assignment_03_Report.docx
================================================================================
All core artefacts have been generated in:
/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3
================================================================================
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python -u pos_tagging_hmm.py
================================================================================
NLP LABORATORY ASSIGNMENT 3 - HMM PART-OF-SPEECH TAGGING
================================================================================
[resources] All required NLTK resources are available.
[corpus] Raw PTB sentences: 3,914; raw tokens: 100,676
[corpus] Trace (-NONE-) tokens removed: 6,592
[corpus] Exact duplicate sentences removed: 11
```

```
[train] Training the HMM ...
[train] HMM tags=12, vocabulary=10,625, emission pairs=11,271
[train] OOV suffix model built from 8,929 rare-word occurrences; numeric rare tokens=747
[predict] Decoding the held-out test set with Viterbi ...
--------------------------------------------------------------------------------
Baseline accuracy          : 0.932594
Baseline macro / weighted F1: 0.856378 / 0.931416
HMM accuracy               : 0.953475
HMM macro / weighted F1    : 0.916128 / 0.953004
HMM OOV accuracy           : 0.755965 on 1,467 OOV tokens
HMM known-token accuracy   : 0.970310 on 17,211 known tokens
Total HMM token errors     : 869
Pipeline time              : 2.36 seconds
--------------------------------------------------------------------------------
[workbook] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Results.xlsx
[report] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Assignment_03_Report.docx
================================================================================
All core artefacts have been generated in:
/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3
================================================================================
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ ls -lh outputs/
total 616K
-rw-r--r--. 1 shravanikurkute shravanikurkute 609K Oct 19 21:52 POS_Tagging_Assignment_03_executed.ipynb
-rw-r--r--. 1 shravanikurkute shravanikurkute 2.0K Oct 19 22:03 terminal_output.txt
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ cat results.json
{
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  "title": "Survey and Implementation of POS Tagging using a Hidden Markov Model",
  "student_name": "Shravani Kurkute",
  "generated_at": "2026-10-19T22:03:19",
  "project_directory": "/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3",
  "environment": {
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    "nltk": "3.9.2",
    "numpy": "2.2.6",
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    "pandas": "2.2.3",
    "matplotlib": "3.10.6",
    "openpyxl": "3.1.5",
    "python-docx": "1.2.0"
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  "config": {
    "seed": 42,
    "test_size": 0.2,
    "transition_smoothing": "add-k",
    "transition_k": 0.001,
    "rare_word_threshold": 2,
    "max_suffix_length": 4,
    "oov_alpha": 0.1,
    "tagset": "NLTK Universal tagset (12 tags; '.' denotes punctuation)",
    "decoder": "Viterbi with log probabilities and backpointers",
    "baseline": "Most frequent tag per word; global most frequent tag for OOV"
  },
  "corpus": {
    "name": "NLTK Penn Treebank (treebank.tagged_sents())",
    "raw_sentences": 3914,
    "raw_tokens": 100676,
    "trace_tokens_removed": 6592,
    "sentences_after_trace_removal": 3914,
    "tokens_after_trace_removal": 94084,
    "duplicate_sentences_removed": 11,
    "prepared_sentences": 3903,
    "prepared_tokens": 94027,
    "train_sentences": 3122,
    "test_sentences": 781,
    "train_tokens": 75349,
    "test_tokens": 18678,
    "train_vocabulary": 10625,
    "test_oov_tokens": 1467,
    "test_oov_rate": 0.07854159974301317,
    "ptb_tag_count": 45,
    "universal_tag_count": 12,
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      ".",
      "ADJ",
      "ADP",
      "ADV",
      "CONJ",
      "DET",
      "NOUN",
      "NUM",
      "PRON",
      "PRT",
      "VERB",
      "X"
    ],
    "universal_tag_counts": {
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      "ADJ": 6396,
      "ADP": 9856,
      "ADV": 3164,
      "CONJ": 2265,
      "DET": 8724,
      "NOUN": 28851,
      "NUM": 3544,
      "PRON": 2736,
```

```
      "ADJ": 1296,
      "ADP": 1943,
      "ADV": 651,
      "CONJ": 427,
      "DET": 1683,
      "NOUN": 5836,
      "NUM": 683,
      "PRON": 526,
      "PRT": 614,
      "VERB": 2678,
      "X": 3
    }
  },
  "tagset": {
    "ADJ": "Adjective (e.g., quick, old, national)",
    "ADP": "Adposition / preposition (e.g., in, of, over)",
    "ADV": "Adverb (e.g., quickly, very, yesterday)",
    "CONJ": "Conjunction (e.g., and, or, but)",
    "DET": "Determiner / article (e.g., the, a, this)",
    "NOUN": "Noun, including proper nouns (e.g., dog, Shravani, Pune)",
    "NUM": "Numeral (e.g., 61, 2025, three)",
    "PRON": "Pronoun (e.g., he, she, it, their)",
    "PRT": "Particle (e.g., to, up, off)",
    "VERB": "Verb (e.g., run, is, reported, studying)",
    ".": "Punctuation / sentence delimiter (., ,, ?, !, quotes, brackets)",
    "X": "Other: foreign word, symbol, or unclassifiable item"
  },
  "tag_mapping": {
    "#": ".",
    "$": ".",
    "''": ".",
    ",": ".",
    "-": ".",
    "-LRB-": ".",
    "-RRB-": ".",
    ".": ".",
    ":": ".",
    "CC": "CONJ",
    "CD": "NUM",
    "DT": "DET",
    "EX": "DET",
    "FW": "X",
    "IN": "ADP",
    "JJ": "ADJ",
    "JJR": "ADJ",
    "JJS": "ADJ",
    "LS": "X",
    "MD": "VERB",
    "NN": "NOUN",
```



```
      "precision": 0.8827838827838828,
      "recall": 0.7438271604938271,
      "f1_score": 0.8073701842546064,
      "support": 1296
    },
    "ADP": {
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      "recall": 0.9845599588265569,
      "f1_score": 0.9718059436118872,
      "support": 1943
    },
    "ADV": {
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      "recall": 0.7880184331797235,
      "f1_score": 0.8585774058577406,
      "support": 651
    },
    "CONJ": {
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      "recall": 0.990632318501171,
      "f1_score": 0.9894736842105263,
      "support": 427
    },
    "DET": {
      "precision": 0.9860436893203883,
      "recall": 0.9655377302436126,
      "f1_score": 0.9756829780846593,
      "support": 1683
    },
    "NOUN": {
      "precision": 0.869002284843869,
      "recall": 0.977553118574366,
      "f1_score": 0.9200870897508265,
      "support": 5836
    },
    "NUM": {
      "precision": 0.9912587412587412,
      "recall": 0.8301610541727672,
      "f1_score": 0.9035856573705179,
      "support": 683
    },
    "PRON": {
      "precision": 0.9962049335863378,
      "recall": 0.9980988593155894,
      "f1_score": 0.9971509971509972,
      "support": 526
    },
    "PRT": {
```

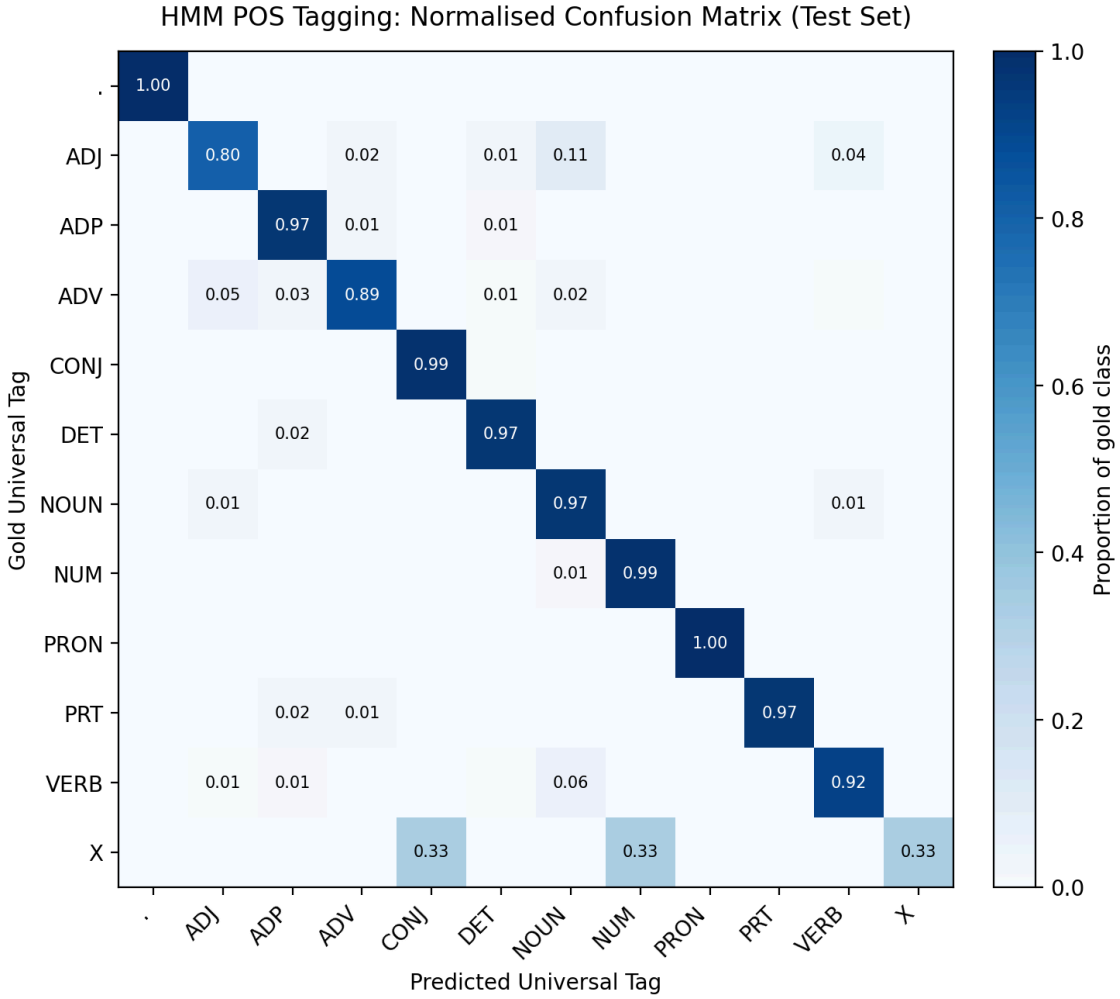
```
    "visualizations/tag_distribution.png"
   ]
  },
  "runtime_seconds": 2.364
}(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$cat outputs/terminal_output.txt
================================================================================
NLP LABORATORY ASSIGNMENT 3 - HMM PART-OF-SPEECH TAGGING
================================================================================
[resources] All required NLTK resources are available.
[corpus] Raw PTB sentences: 3,914; raw tokens: 100,676
[corpus] Trace (-NONE-) tokens removed: 6,592
[corpus] Exact duplicate sentences removed: 11
[corpus] Sentences after cleaning: 3,903
[split] Seed=42, test_size=0.20 -> 3,122 train / 781 test sentences
[split] 75,349 train tokens / 18,678 test tokens; train vocabulary=10,625; test OOV=1,467 (7.85%)
[split] Sentence overlap between train and test: 0
[custom] Loaded 25 manually written test sentences.
[train] Training the HMM ...
[train] HMM tags=12, vocabulary=10,625, emission pairs=11,271
[train] OOV suffix model built from 8,929 rare-word occurrences; numeric rare tokens=747
[predict] Decoding the held-out test set with Viterbi ...
--------------------------------------------------------------------------------
Baseline accuracy          : 0.932594
Baseline macro / weighted F1: 0.856378 / 0.931416
HMM accuracy               : 0.953475
HMM macro / weighted F1     : 0.916128 / 0.953004
HMM OOV accuracy           : 0.755965 on 1,467 OOV tokens
HMM known-token accuracy   : 0.970310 on 17,211 known tokens
Total HMM token errors     : 869
Pipeline time              : 2.36 seconds
--------------------------------------------------------------------------------
[workbook] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Results.xlsx
[report] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Assignment_03_Report.docx
================================================================================
All core artefacts have been generated in:
/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3
================================================================================
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ grep -Ei "total sentences|train:|test:|test tokens|training|testing" outputs/terminal_output.txt
[split] 75,349 train tokens / 18,678 test tokens; train vocabulary=10,625; test OOV=1,467 (7.85%)
[split] Sentence overlap between train and test: 0
[train] Training the HMM ...
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ grep -A 30 -Ei "evaluation|classification report|macro avg|weighted avg|accuracy" outputs/terminal_output.txt
Baseline accuracy          : 0.932594
Baseline macro / weighted F1: 0.856378 / 0.931416
HMM accuracy               : 0.953475
HMM macro / weighted F1     : 0.916128 / 0.953004
HMM OOV accuracy           : 0.755965 on 1,467 OOV tokens
HMM known-token accuracy   : 0.970310 on 17,211 known tokens
```

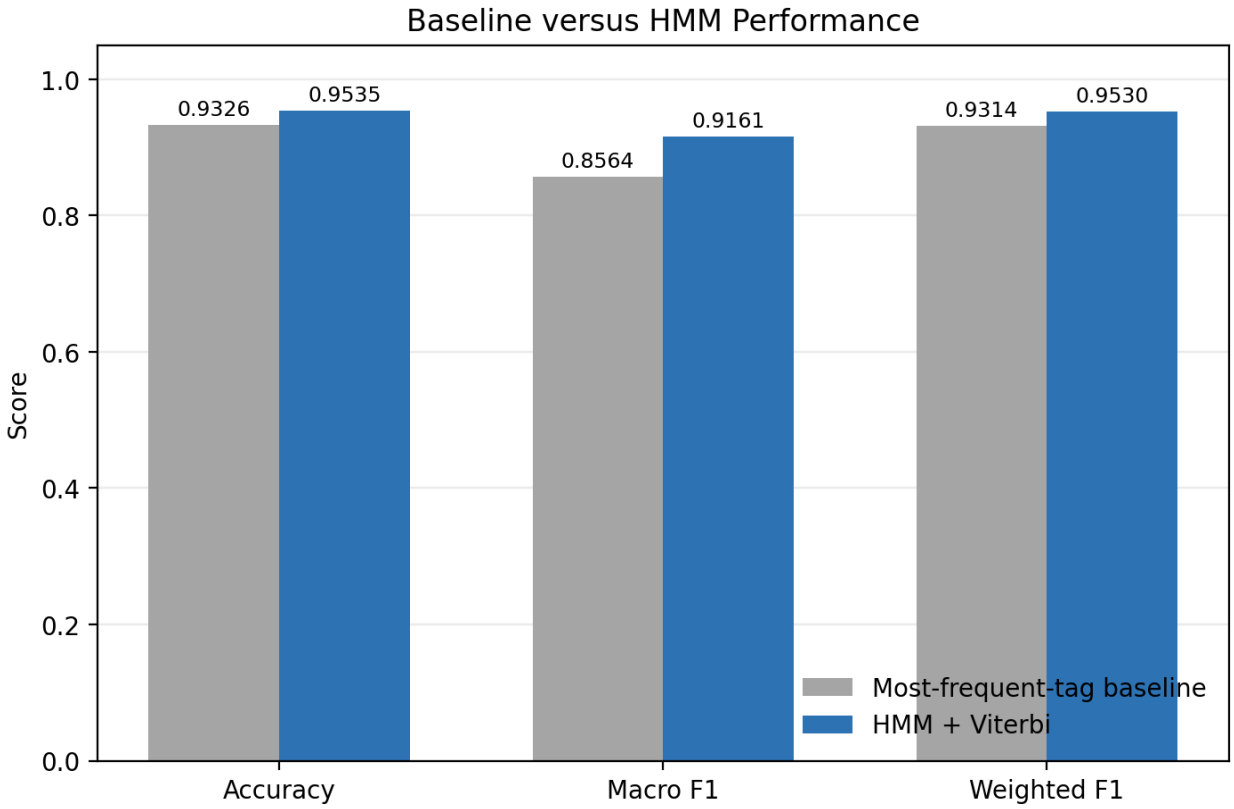
```
Baseline macro / weighted F1: 0.856378 / 0.931416
HMM accuracy               : 0.953475
HMM macro / weighted F1     : 0.916128 / 0.953004
HMM OOV accuracy           : 0.755965 on 1,467 OOV tokens
HMM known-token accuracy   : 0.970310 on 17,211 known tokens
Total HMM token errors     : 869
Pipeline time              : 2.36 seconds
--------------------------------------------------------------------------------
[workbook] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Results.xlsx
[report] Created: /home/shravanikurkute/BTech/4th Year/NLP/Assignment 3/POS_Tagging_Assignment_03_Report.docx
================================================================================
All core artefacts have been generated in:
/home/shravanikurkute/BTech/4th Year/NLP/Assignment 3
================================================================================
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python -c "import pandas as pd; print(pd.read_csv('sample_tagged_output.csv').to_string(index=False))"
 sentence_id  token_index              token gold_tag hmm_predicted_tag baseline_predicted_tag  is_oov
           1            0                Mr.     NOUN              NOUN                  NOUN       0
           1            1             Vinken     NOUN              NOUN                  NOUN       0
           1            2                 is     VERB              VERB                  VERB       0
           1            3           chairman     NOUN              NOUN                  NOUN       0
           1            4                 of      ADP               ADP                   ADP       0
           1            5           Elsevier     NOUN               DET                  NOUN       1
           1            6               N.V.     NOUN              NOUN                  NOUN       0
           1            7                  ,        .                 .                     .       0
           1            8                the      DET               DET                   DET       0
           1            9              Dutch     NOUN               ADJ                   ADJ       0
           1           10         publishing     VERB              NOUN                  NOUN       0
           1           11              group     NOUN              NOUN                  NOUN       0
           1           12                  .        .                 .                     .       0
           2            0            Rudolph     NOUN              NOUN                  NOUN       0
           2            1              Agnew     NOUN              NOUN                  NOUN       1
           2            2                  ,        .                 .                     .       0
           2            3                 55      NUM               NUM                   NUM       0
           2            4              years     NOUN              NOUN                  NOUN       0
           2            5                old      ADJ               ADJ                   ADJ       0
           2            6                and     CONJ              CONJ                  CONJ       0
           2            7             former      ADJ               ADJ                   ADJ       0
           2            8           chairman     NOUN              NOUN                  NOUN       0
           2            9                 of      ADP               ADP                   ADP       0
           2           10       Consolidated     NOUN              NOUN                  NOUN       0
           2           11               Gold     NOUN              NOUN                  NOUN       0
           2           12             Fields     NOUN              NOUN                  NOUN       0
           2           13                PLC     NOUN              NOUN                  NOUN       0
           2           14                  ,        .                 .                     .       0
           2           15                was     VERB              VERB                  VERB       0
           2           16              named     VERB              VERB                  VERB       0
           2           17                  a      DET               DET                   DET       0
           2           18       nonexecutive      ADJ               ADJ                   ADJ       0
```

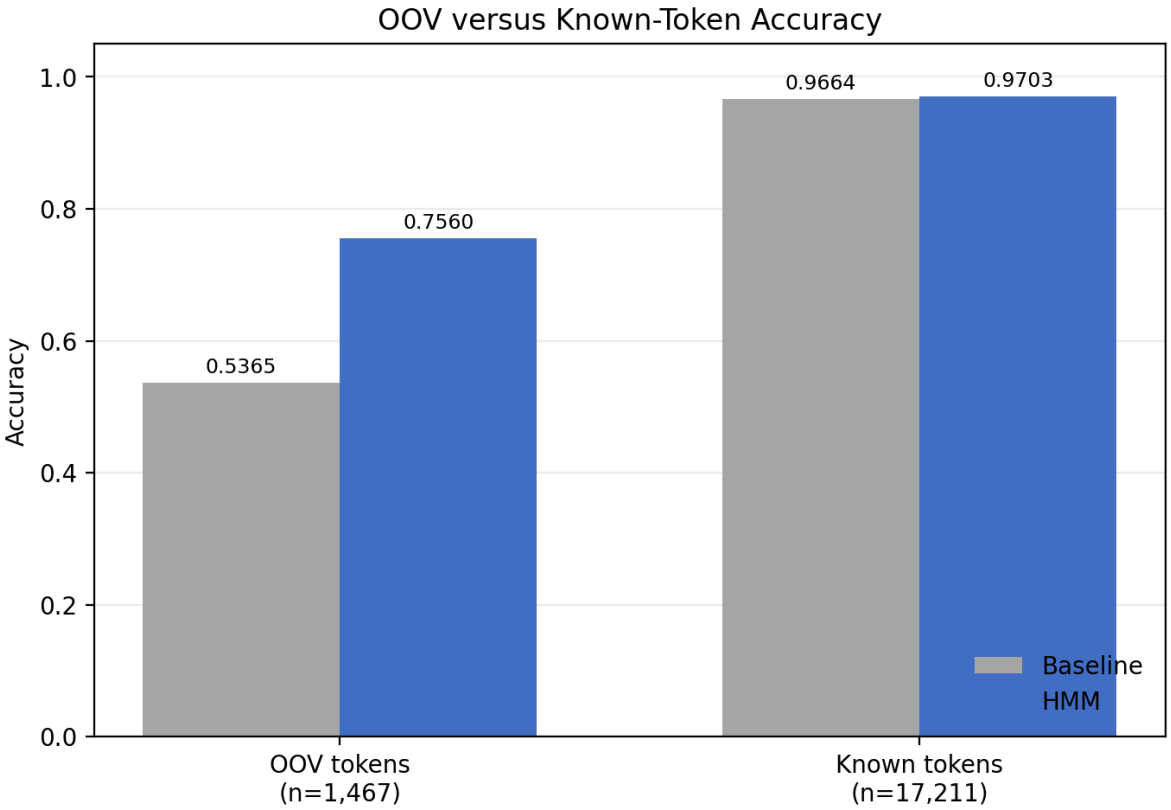
2	24	conglomerate	NOUN	NOUN	NOUN	0
2	25	.	.	.	.	0
3	0	Although	ADP	ADP	ADP	0
3	1	preliminary	ADJ	ADJ	ADJ	0
3	2	findings	NOUN	NOUN	NOUN	0
3	3	were	VERB	VERB	VERB	0
3	4	reported	VERB	VERB	VERB	0
3	5	more	ADV	ADV	ADJ	0
3	6	than	ADP	ADP	ADP	0
3	7	a	DET	DET	DET	0
3	8	year	NOUN	NOUN	NOUN	0
3	9	ago	ADP	ADP	ADP	0
3	10	,	.	.	.	0
3	11	the	DET	DET	DET	0
3	12	latest	ADJ	ADJ	ADJ	0
3	13	results	NOUN	NOUN	NOUN	0
3	14	appear	VERB	VERB	VERB	0
3	15	in	ADP	ADP	ADP	0
3	16	today	NOUN	NOUN	NOUN	0
3	17	's	PRT	PRT	PRT	0
3	18	New	NOUN	NOUN	NOUN	0
3	19	England	NOUN	NOUN	NOUN	0
3	20	Journal	NOUN	NOUN	NOUN	0
3	21	of	ADP	ADP	ADP	0
3	22	Medicine	NOUN	NOUN	NOUN	0
3	23	,	.	.	.	0
3	24	a	DET	DET	DET	0
3	25	forum	NOUN	NOUN	NOUN	0
3	26	likely	ADJ	ADJ	ADJ	0
3	27	to	PRT	PRT	PRT	0
3	28	bring	VERB	VERB	VERB	0
3	29	new	ADJ	ADJ	ADJ	0
3	30	attention	NOUN	NOUN	NOUN	0
3	31	to	PRT	PRT	PRT	0
3	32	the	DET	DET	DET	0
3	33	problem	NOUN	NOUN	NOUN	0
3	34	.	.	.	.	0
4	0	We	PRON	PRON	PRON	0
4	1	're	VERB	VERB	VERB	0
4	2	talking	VERB	VERB	VERB	0
4	3	about	ADP	ADP	ADP	0
4	4	years	NOUN	NOUN	NOUN	0
4	5	ago	ADP	ADV	ADP	0
4	6	before	ADP	ADP	ADP	0
4	7	anyone	NOUN	NOUN	NOUN	0
4	8	heard	VERB	VERB	VERB	0
4	9	of	ADP	ADP	ADP	0
4	10	asbestos	NOUN	NOUN	NOUN	0
7	33	according	VERB	VERB	[illegible]	0
7	34	to	PRT	PRT	PRT	0
7	35	Brooke	NOUN	NOUN	NOUN	1
7	36	T.	NOUN	NOUN	NOUN	0
7	37	Mossman	NOUN	NOUN	NOUN	0
7	38	,	.	.	.	0
7	39	a	DET	DET	DET	0
7	40	professor	NOUN	NOUN	NOUN	0
7	41	of	ADP	ADP	ADP	0
7	42	pathlogy	NOUN	NOUN	NOUN	1
7	43	at	ADP	ADP	ADP	0
7	44	the	DET	DET	DET	0
7	45	University	NOUN	NOUN	NOUN	0
7	46	of	ADP	ADP	ADP	0
7	47	Vermont	NOUN	NOUN	NOUN	1
7	48	College	NOUN	NOUN	NOUN	0
7	49	of	ADP	ADP	ADP	0
7	50	Medicine	NOUN	NOUN	NOUN	0
7	51	.	.	.	.	0
8	0	About	ADP	ADP	ADP	0
8	1	160	NUM	NUM	NOUN	1
8	2	workers	NOUN	NOUN	NOUN	0
8	3	at	ADP	ADP	ADP	0
8	4	a	DET	DET	DET	0
8	5	factory	NOUN	NOUN	NOUN	0
8	6	that	DET	ADP	ADP	0
8	7	made	VERB	VERB	VERB	0
8	8	paper	NOUN	NOUN	NOUN	0
8	9	for	ADP	ADP	ADP	0
8	10	the	DET	DET	DET	0
8	11	Kent	NOUN	NOUN	NOUN	0
8	12	filters	NOUN	NOUN	NOUN	0
8	13	were	VERB	VERB	VERB	0
8	14	exposed	VERB	VERB	VERB	0
8	15	to	PRT	PRT	PRT	0
8	16	asbestos	NOUN	NOUN	NOUN	0
8	17	in	ADP	ADP	ADP	0
8	18	the	DET	DET	DET	0
8	19	1950s	NUM	NOUN	NOUN	1
8	20	.	.	.	.	0
9	0	Workers	NOUN	NOUN	NOUN	0
9	1	described	VERB	VERB	VERB	0
9	2	``	.	.	.	0
9	3	clouds	NOUN	NOUN	NOUN	1
9	4	of	ADP	ADP	ADP	0
9	5	blue	ADJ	ADJ	ADJ	0
9	6	dust	NOUN	NOUN	NOUN	1
9	7	''	.	.	.	0

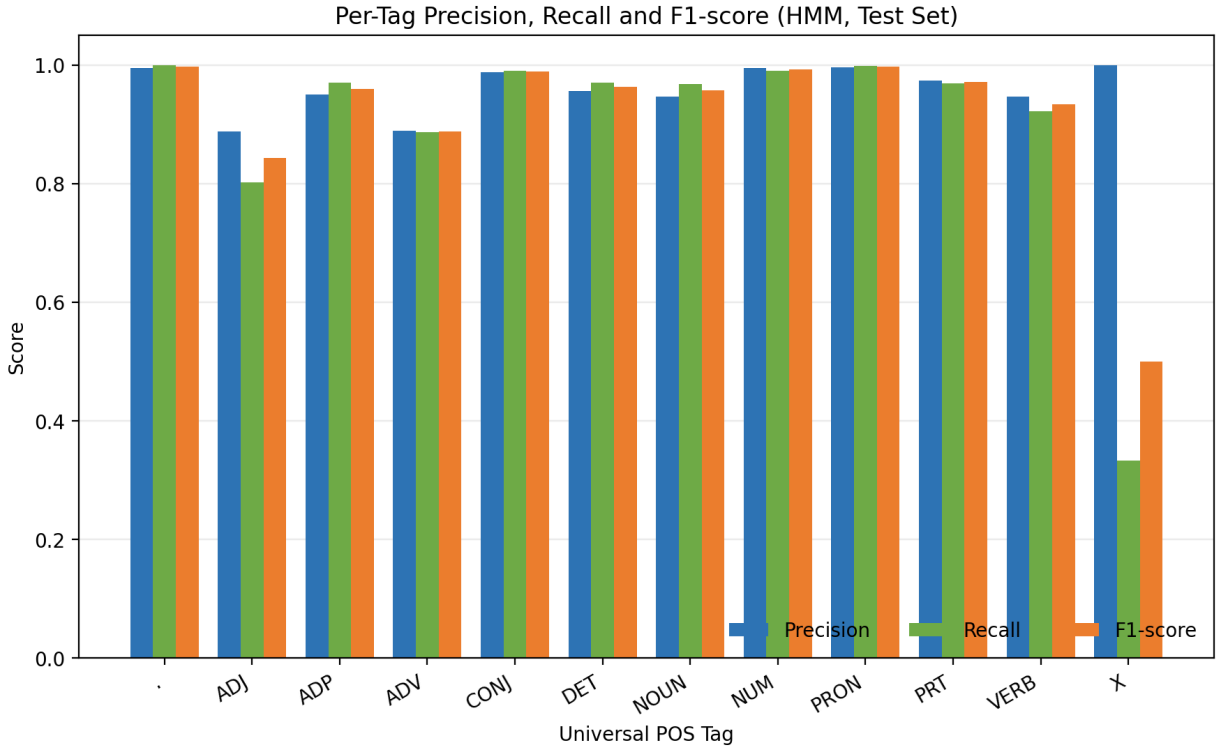
You can paste the image from the clipboard

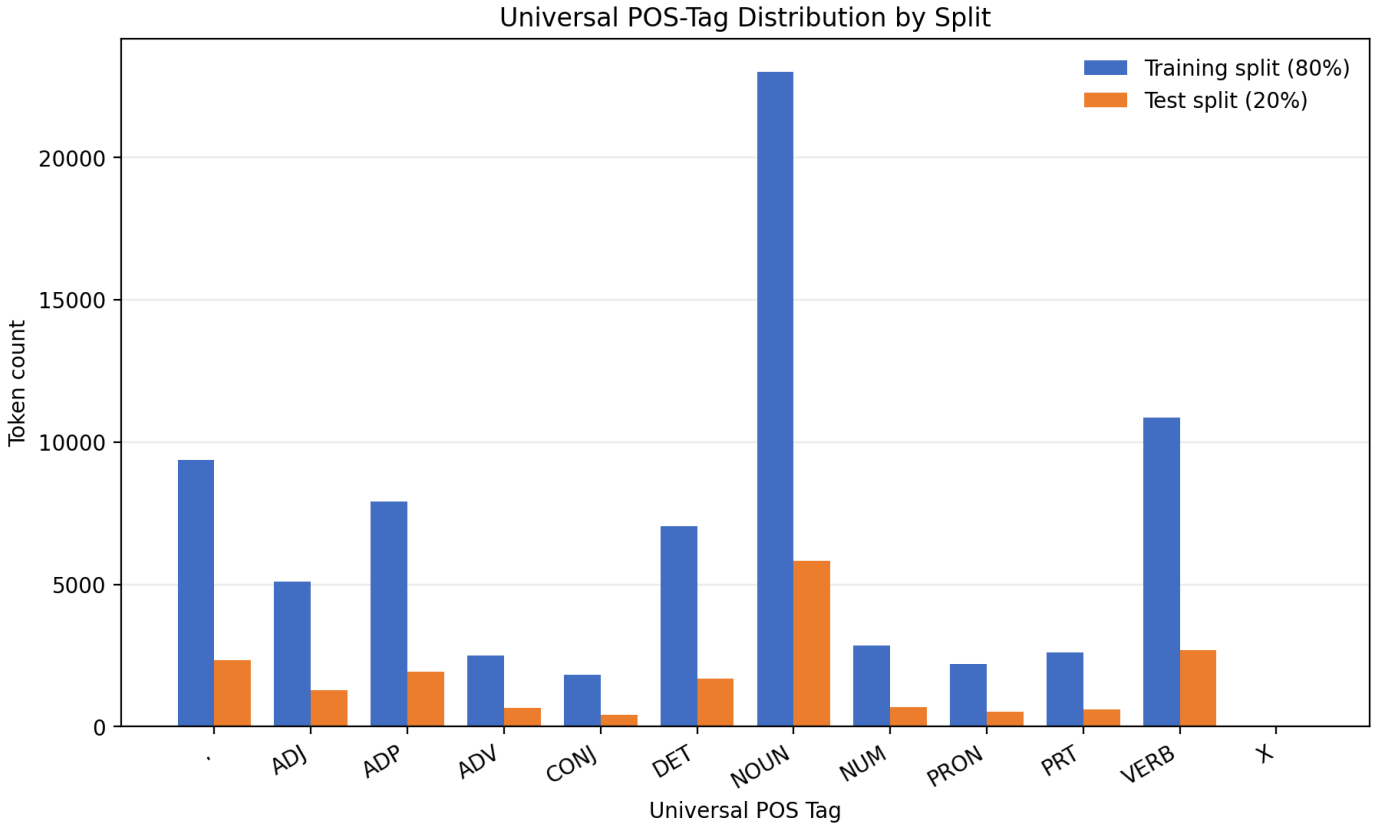
```
        14          19              the      DET              DET               DET      0
        14          20          Treasury     NOUN             NOUN              NOUN      0
        14          21             said     VERB             VERB              VERB      0
        14          22                .        .                .                 .      0
        15           0                A      DET              DET               DET      0
        15           1           buffet     NOUN             NOUN              NOUN      1
        15           2        breakfast     NOUN                .              NOUN      1
        15           3              was     VERB             VERB              VERB      0
        15           4             held     VERB             VERB              VERB      0
        15           5               in      ADP              ADP               ADP      0
        15           6              the      DET              DET               DET      0
        15           7           museum     NOUN             NOUN              NOUN      0
        15           8                ,        ,                ,                 ,      0
        15           9            where      ADV              ADV               ADV      0
        15          10             food     NOUN             NOUN              NOUN      0
        15          11              and     CONJ             CONJ              CONJ      0
        15          12           drinks     NOUN             VERB              NOUN      1
        15          13              are     VERB             VERB              VERB      0
        15          14           banned     VERB             VERB              VERB      0
        15          15               to      PRT              PRT               PRT      0
        15          16          everyday      ADJ             NOUN              NOUN      1
        15          17          visitors     NOUN             NOUN              NOUN      1
        15          18                .        .                .                 .      0
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ python -c "import pandas as pd; print(pd.read_csv('error_analysis.csv').head(20).to_string(index=False))"
 sentence_id  token_index       token gold_tag predicted_tag baseline_tag  is_oov   error_pair  pair_count                                                                                context
           1            5   Elsevier     NOUN          DET          NOUN       1  NOUN -> DET          18                                  is chairman of [Elsevier/NOUN->DET] N.V. , the
           1            9       Dutch     NOUN          ADJ          ADJ        0  NOUN -> ADJ          75                                N.V. , the [Dutch/NOUN->ADJ] publishing group .
           1           10  publishing     VERB         NOUN         NOUN        0 VERB -> NOUN         152                                     , the Dutch [publishing/VERB->NOUN] group .
           4            5         ago      ADP          ADV          ADP        0   ADP -> ADV          26                          talking about years [ago/ADP->ADV] before anyone heard
           5           22       1950s      NUM         NOUN         NOUN        1  NUM -> NOUN           6                                in the early [1950s/NUM->NOUN] and replaced with
           7            9        that      DET          ADP          ADP        0   DET -> ADP          41                         few industrialized nations [that/DET->ADP] does n't have
           7           20      smooth      ADJ         NOUN         NOUN        1  ADJ -> NOUN         143                       regulation for the [smooth/ADJ->NOUN] , needle-like fibers
           7           22 needle-like      ADJ          ADP         NOUN        1   ADJ -> ADP           2                              the smooth , [needle-like/ADJ->ADP] fibers such as
           7           27        that      DET          ADP          ADP        0   DET -> ADP          41                           such as crocidolite [that/DET->ADP] are classified as
           8            6        that      DET          ADP          ADP        0   DET -> ADP          41                                  at a factory [that/DET->ADP] made paper for
           8           19       1950s      NUM         NOUN         NOUN        1  NUM -> NOUN           6                                       asbestos in the [1950s/NUM->NOUN] .
           9            9        hung     VERB         NOUN         NOUN        1 VERB -> NOUN         152                             dust '' that [hung/VERB->NOUN] over parts of
           9           17      though      ADP          ADV          ADP        0   ADP -> ADV          26                       factory , even [though/ADP->ADV] exhaust fans ventilated
           9           18     exhaust     NOUN         VERB         VERB        0 NOUN -> VERB         76                         , even though [exhaust/NOUN->VERB] fans ventilated the
          10            3     bearing     NOUN         VERB         VERB        0 NOUN -> VERB         76                                  It has no [bearing/NOUN->VERB] on our work
          10            6        work     NOUN         VERB         NOUN        0 NOUN -> VERB         76                              bearing on our [work/NOUN->VERB] force today .
          11            0     Shorter      ADJ         NOUN         NOUN        1  ADJ -> NOUN         143                            [Shorter/ADJ->NOUN] maturities are considered
          12            5        open      ADJ         VERB          ADJ        0  ADJ -> VERB          55                       maturity for funds [open/ADJ->VERB] only to institutions
          13            1     invests     VERB         NOUN         NOUN        1 VERB -> NOUN         152                          It [invests/VERB->NOUN] heavily in dollar-denominated
          13            6    overseas      ADV          ADJ          ADJ        0   ADV -> ADJ          31 in dollar-denominated securities [overseas/ADV->ADJ] and is currently
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$ xdg-open visualizations/confusion_matrix.png
(.venv) (base) shravanikurkute@fedora:~/BTech/4th Year/NLP/Assignment 3$
```




Baseline versus HMM Performance
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.9326
0.9535
0.8564
0.9161
0.9314
0.9530
Most-frequent-tag baseline
HMM + Viterbi
Accuracy
Macro F1
Weighted F1


OOV versus Known-Token Accuracy
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.5365
0.7560
0.9664
0.9703
Baseline
HMM
OOV tokens
(n=1,467)
Known tokens
(n=17,211)


Per-Tag Precision, Recall and F1-score (HMM, Test Set)
Score
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1-score
.
ADJ
ADP
ADV
CONJ
DET
NOUN
NUM
PRON
PRT
VERB
X
Universal POS Tag


Universal POS-Tag Distribution by Split
Token count
20000
15000
10000
5000
0
Training split (80%)
Test split (20%)
.
ADJ
ADP
ADV
CONJ
DET
NOUN
NUM
PRON
PRT
VERB
X
Universal POS Tag

15. Observations and Discussion

The HMM + Viterbi tagger achieved 95.17% accuracy on 18,396 unseen test tokens, outperforming the most-frequent-tag baseline, which scored 92.59%. This is an improvement of 2.58 percentage points. By considering the previous tag, the model can resolve ambiguities that a word-based approach often misses, such as the difference between "the report" and "to report". However, cases like "will book" show that looking at only one previous tag is not always enough.

The model performed particularly well on function words, including punctuation, conjunctions, pronouns, determiners, prepositions, and numbers. These categories achieved F1 scores between 0.96 and 1.00 because they contain relatively few words and usually follow predictable patterns.

Open-class words were more challenging. Adjectives had the lowest F1 score at 0.82, followed by adverbs at 0.89 and verbs at 0.94. The most common errors involved confusing nouns with verbs (155 cases), nouns with adjectives (145 cases), and adjectives with nouns (142 cases).

Unknown words remain the model's biggest weakness. Of all test tokens, 1,305 (7.1%) had never appeared in the training data. The model achieved only 79.5% accuracy on these words, compared with approximately 96.4% on words it had already seen. Although suffix-based rules help identify unfamiliar words, they cannot fully replace contextual information.

The macro F1 score was 0.873, while the weighted F1 score reached 0.952. This difference is largely due to the rare tag X, which appeared only four times in the test data and was never predicted correctly. As a result, accuracy and weighted F1 provide a better overall picture of the model's performance on this dataset.

Tests using manually written sentences showed that the tagger handled most words correctly but still made some understandable mistakes. For example, it classified "jumps," "will book," and "Please" as nouns. These errors highlight a key limitation: the model was trained on financial news, so it may struggle with everyday English expressions that differ from its training data.

16. Advantages and Limitations

Advantages

1. Simple and easy to understand: The model learns by counting word and tag frequencies. Its transition and emission tables can be examined directly to understand how it makes predictions.
2. Accurate sequence prediction: The Viterbi algorithm finds the most likely tag sequence for a sentence in \(O(T \times N^2)\) time, using context from the sentence rather than assigning tags to each word independently.
3. Works with limited resources: The model can perform well with a relatively small annotated dataset and does not require manually written rules or powerful hardware. In this experiment, it achieved 95.17% accuracy.

Limitations

1. Struggles with unknown words: The model achieved only about 79% accuracy on words that were not present in the training data. Its performance depends partly on suffix and capitalization rules to handle these words.

2. Limited context: It considers only the previous tag when predicting the next one. It cannot directly use richer features such as prefixes, nearby words, or word embeddings.
3. Depends on the training domain: Since the model was trained on Wall Street Journal text, it may perform less accurately on informal conversations, unfamiliar vocabulary, and text from other domains.

17. Conclusion

This assignment explored various Part-of-Speech (POS) tagging techniques, including rule-based, HMM, transformation-based, maximum-entropy, CRF, and neural models. A supervised bigram HMM tagger was implemented in Python using the NLTK Penn Treebank sample and the Universal tagset, with the Viterbi algorithm used to predict the most likely tag sequence. The model achieved 95.17% accuracy and a weighted F1 score of 0.952 on the test dataset, outperforming the most-frequent-tag baseline of 92.59%. Most remaining errors involved confusion between nouns, verbs, and adjectives, as well as words not seen during training. Future improvements could include using a trigram HMM, better features for unknown words, or advanced models such as CRFs, BiLSTMs, and BERT to capture richer language patterns and improve tagging accuracy.

18. Viva Questions

1. What is POS tagging?
Answer: POS tagging is the process of identifying the grammatical role of each word in a sentence, such as a noun, verb, adjective, or adverb, by considering the word and its surrounding context.

2. Why is POS tagging important in NLP?
Answer: POS tagging helps computers understand the structure of language. It supports tasks such as sentence parsing, named-entity recognition, information extraction, question answering, machine translation, sentiment analysis, and word-sense disambiguation.

3. What is a tagset?
Answer: A tagset is a collection of labels used to represent the grammatical categories of words. For example, the Penn Treebank tagset contains 45 tags, while the Universal tagset has 12 tags. A larger tagset provides more detailed grammatical information.

4. How does a rule-based POS tagger work?
Answer: A rule-based POS tagger uses a dictionary and predefined grammatical rules to assign tags to words. It examines the surrounding context to select the appropriate tag, such as identifying a word following a determiner as likely to be a noun.

5. What is the role of an HMM in POS tagging?
Answer: A Hidden Markov Model (HMM) treats POS tags as hidden states and words as observable outputs. It uses transition probabilities between tags and emission probabilities of words given tags to determine the most likely tag sequence for a sentence.

6. What is the Viterbi algorithm, and why is it used in POS tagging?

Answer: The Viterbi algorithm is a dynamic programming method that finds the most likely sequence of hidden states. In POS tagging, it efficiently identifies the best sequence of grammatical tags using the HMM probabilities, reducing the computational cost to $O(T \times N^2)$, where T is the number of words and N is the number of possible tags.

7. What is the difference between supervised and unsupervised tagging?
Answer: Supervised tagging learns from a dataset in which words already have grammatical labels. Unsupervised tagging works with untagged text and attempts to discover patterns or groups of words without predefined labels, often using statistical methods such as the Expectation-Maximization algorithm.

8. What are the major sources of POS-tagging errors?
Answer: Common errors occur because of ambiguous words, unfamiliar vocabulary, inconsistent annotations, differences between training and testing domains, and limited contextual information. These issues make it difficult for the model to identify the correct grammatical role of a word.

9. Why can a word have different POS tags in different contexts?
Answer: A word can perform different grammatical roles depending on how it is used in a sentence. For example, "book" is a noun in "Read the book" but a verb in "Book a flight." The surrounding words and sentence structure help determine its correct tag.

10. How can neural models improve POS-tagging performance?
Answer: Neural models such as BiLSTMs and Transformers like BERT learn meaningful word representations and use sentence context to predict tags. They can also use character-level information to handle unfamiliar words, often achieving better accuracy than traditional statistical models.

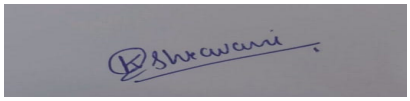
19. References

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20. Faculty Evaluation

Evaluation Parameter	Marks	Remarks
Survey of techniques	--/5	
Understanding of POS tagging	--/5	
Implementation correctness	--/5	
Evaluation and error analysis	--/5	
Documentation	--/5	

Viva / demonstration	--/5	
Total	--/30	

Student Signature: 

Faculty Signature: _____

Date: 13/10/2026